

# Book XI of Euclid's Elements

Formal Reconstruction — Technical Documentation

Jurek Róžański

Military University of Technology, Warsaw, Poland

CGJTEAMLAB

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# Chapter 1

## Proposition XI.4

### 1.1 Statement

Euclid XI.4 gives the basic criterion for perpendicularity to a plane: if a line through the common point of two intersecting lines of a plane is perpendicular to each of those two lines, then it is perpendicular to the plane itself.

The formal reconstruction is implemented in

```
CGJteamLab/Proposition11_4.lean
```

on top of the spatial infrastructure in

```
CGJteamLab/Hilbert3DAxioms.lean  
CGJteamLab/Hilbert3DInterface.lean
```

The Lean source is authoritative for theorem signatures, helper names, actual dependencies, and proof status.

### 1.2 Euclid XI.3 and the target notion

Book XI, Definition 3 says that a straight line is perpendicular to a plane when it makes right angles with every straight line which meets it and lies in that plane. The project encodes exactly this universal synthetic notion.

For two incidence lines:

```
def HilbertLinesPerpendicularAt  
  [H : HilbertIncidence Geo]  
  (l m : Geo.Line)  
  (O : Geo.Point) : Prop :=  
  H.OnLine O l /\  
  H.OnLine O m /\  
  exists A B : Geo.Point,  
    Ne A O /\  
    Ne B O /\
```

```

H.OnLine A l /\
H.OnLine B m /\
Not (PrimCollinear Geo A O B) /\
HilbertRightAngle Geo A O B

```

The explicit noncollinearity condition is important. It prevents a degenerate “self-perpendicular” witness and yields the reusable theorem

```

hilbert_linesPerpendicularAt_ne

```

showing that perpendicular lines are distinct.

For a line and a plane:

```

def HilbertLinePerpendicularPlaneAt
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l : Geo.Line)
  (pi : S.Plane)
  (O : Geo.Point) : Prop :=
H.OnLine O l /\
S.OnPlane O pi /\
forall m : Geo.Line,
  HilbertLineInPlane Geo m pi ->
  H.OnLine O m ->
  HilbertLinesPerpendicularAt Geo l m O

```

Thus the conclusion of XI.4 is not a metric abbreviation. It is literally the universal quantification required by Euclid XI.Def.3.

### 1.3 Classical configuration

Let  $m, n$  be two distinct straight lines of a plane  $\pi$ , meeting at  $O$ , and let  $l$  be a line through  $O$  perpendicular to both.

The desired conclusion is that  $l$  is perpendicular to every line  $q \subset \pi$  through  $O$ .

### 1.4 Euclid’s classical proof

Euclid makes the argument constructive.

Take equal segments on the two given plane lines:

$$AE = EB = CE = ED,$$

where  $E$  is the common point of the two given lines. Draw an arbitrary straight line  $GEH$  through  $E$  in the plane, choose an arbitrary point  $F$  on the elevated perpendicular, and join the auxiliary segments.

The classical dependency chain is:

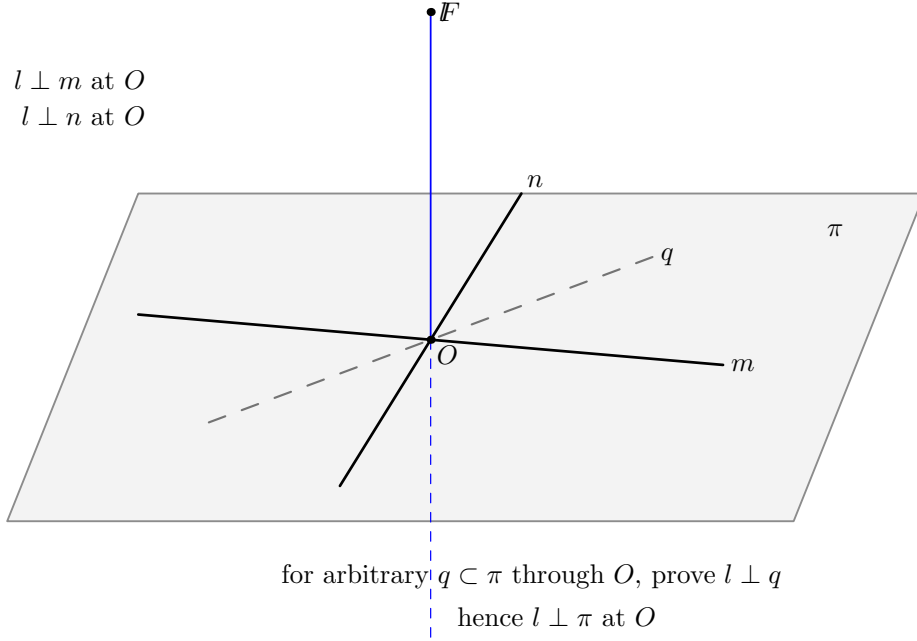


Figure 1.1: Classical configuration for Euclid XI.4. The ambient line  $l$  passes through  $O$  and is perpendicular at  $O$  to two distinct lines  $m, n$  lying in the plane  $\pi$ . The dashed line  $q$  represents an arbitrary line of  $\pi$  through  $O$ , which is the universal target of the conclusion.

1. Vertical angles at  $E$  are equal (I.15). Together with the four equal radial segments, I.4 gives

$$AD = CB$$

and a corresponding angle equality.

2. Applying I.26 to the planar transversal gives

$$GE = EH, \quad AG = BH.$$

3. Since  $FE$  is perpendicular to both original plane lines, I.4 applied to two right triangles gives

$$FA = FB, \quad FC = FD.$$

4. I.8 compares two spatial triangles and supplies the angle relation needed to transport the planar data to the arbitrary transversal.

5. A further I.4 gives

$$FG = FH.$$

6. Finally, from

$$GE = EH, \quad FE = FE, \quad FG = FH,$$

I.8 yields

$$\angle GEF = \angle HEF.$$

Since  $G - E - H$ , the adjacent equal angles are right angles.

7. The transversal  $GEH$  was arbitrary. Hence  $FE$  makes a right angle with every line through  $E$  in the plane. By XI.Def.3,  $FE \perp \pi$ .

This is a genuinely three-dimensional use of Book I congruence theory: several compared triangles are not assumed to lie in one common plane.

## 1.5 Formal theorem

The production theorem is:

```

theorem euclid_proposition_11_4
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (pi : S.Plane)
  (m n : PlaneLine Geo pi)
  (l : Geo.Line)
  (O : PlanePoint Geo pi)
  (hmn : Ne m n)
  (hPerpM :
    HilbertLinesPerpendicularAt Geo l m.1 O.1)
  (hPerpN :
    HilbertLinesPerpendicularAt Geo l n.1 O.1) :
  HilbertLinePerpendicularPlaneAt Geo l pi O.1

```

The theorem does not assume separately that  $l \neq m$  or  $l \neq n$ . Those facts follow from the strengthened nondegenerate line–line perpendicularity predicate.

## 1.6 Spatial architecture

### 1.6.1 No ambient planar type-class instance

The ambient geometry is three-dimensional. The proof deliberately does *not* install a global instance

```
HilbertCongruence Geo
```

or

```
HilbertOrder Geo
```

on the whole space. Planar reasoning is performed only in explicit plane slices

$$\text{PlaneGeo}(\pi).$$

The corresponding points and lines are subtypes:

```

PlanePoint Geo pi
PlaneLine   Geo pi

```



and the interface provides bridges between induced planar facts and ambient facts, for example

```
planeGeo_congruent
planeGeo_angleCongruent_iff_ambient
planeGeo_sameRay_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_linesPerpendicularAt_iff_ambient
```

This separation is one of the central architectural features of the proof.

### 1.6.2 Spatial congruence

Euclid freely applies I.4 and I.8 to triangles in different planes. The project reconstructs this through the spatial congruence layer rather than by coordinates or by an ambient Euclidean metric.

The general spatial infrastructure includes:

```
hilbert_space_congruent_reflexive
hilbert_space_congruent_symmetry
hilbert_space_sas_remaining_angles
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

The target triangle may be placed in its own explicit plane and compared with an ambient source triangle. This is the formal counterpart of Euclid's unqualified use of plane triangle congruence inside solid geometry.

## 1.7 The balanced cross

The first construction is the formal analogue of Euclid's

$$AE = EB = CE = ED.$$

The helper

```
planeGeo_opposite_points_on_line_congruent_to
```

constructs opposite equal points on one line. Applying it twice yields

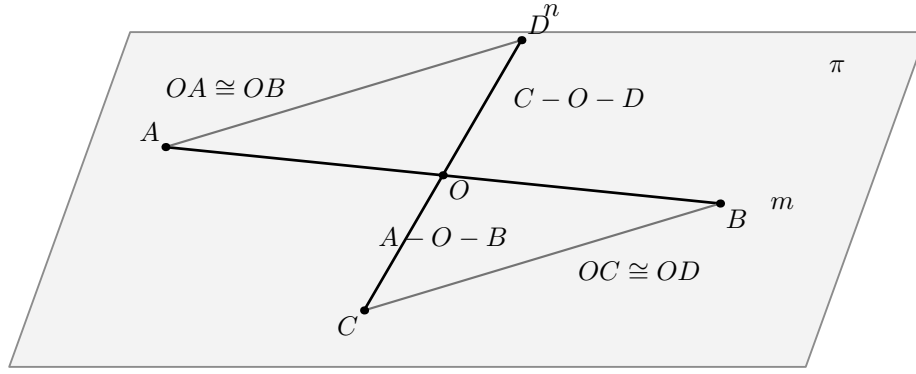
```
hilbert_XI4_balanced_cross_exists
```

and points  $A, B, C, D$  satisfying

$$A - O - B, \quad C - O - D$$

and

$$OA \cong OR, \quad OB \cong OR, \quad OC \cong OR, \quad OD \cong OR.$$



all objects in this figure lie in  $\pi$

Figure 1.2: Balanced cross in  $\pi$ . Points  $A, B$  lie on  $m$ , points  $C, D$  lie on  $n$ , with  $A - O - B$ ,  $C - O - D$ ,  $OA \cong OB$ , and  $OC \cong OD$ . The connector segments  $AD$  and  $BC$  are the first pair used by Euclid's congruence argument.

The first triangle comparison is isolated by

```
planeGeo_XI4_cross_noncollinear
hilbert_XI4_balanced_cross_first_SAS
```

and proves the ambient consequences

$$AD \cong BC, \quad \angle OAD \cong \angle OBC.$$

This is the formal Hilbert version of Euclid's first use of I.4.

## 1.8 Fixed-transversal prototype

Before the proof was generalized to every transversal, a fixed connector configuration was developed. Its role is mathematically transparent: given

$$A - G - D, \quad B - J - C, \quad G - O - J,$$

the balanced cross implies

$$OG \cong OJ, \quad AG \cong BJ.$$

The relevant planar ingredients are

```
planeGeo_XI4_transversal_noncollinear
hilbert_XI4_transversal_ASA
```

corresponding to Euclid's I.26 step.

The spatial point  $F \in l$ , together with the two original perpendicularities, then gives

$$AF \cong BF, \quad CF \cong DF.$$

The proof packages this in

```

hilbert_XI4_spatial_perpendicular_bisector_equidistant
hilbert_XI4_spatial_opposite_pairs_equidistant

```

The spatial SSS/SAS chain is then:

```

hilbert_XI4_spatial_SSS_angle
hilbert_XI4_space_angle_sameRay_first_congruent
hilbert_XI4_spatial_SSS_angle_normalized
hilbert_XI4_spatial_SAS_transversal_side
hilbert_XI4_spatial_transversal_F_equidistant
hilbert_XI4_second_spatial_SSS_angle
hilbert_XI4_right_angle_of_adjacent_equal

```

Its mathematical content is

$$\begin{aligned}
& AD \cong BC, \quad AF \cong BF, \quad CF \cong DF \\
& \quad \Downarrow \\
& \quad \angle DAF \cong \angle CBF \\
& \quad \Downarrow \\
& \quad A - G - D, \quad B - J - C \\
& \quad \Downarrow \\
& \quad \angle GAF \cong \angle JBF \\
& \quad \Downarrow \\
& \quad AG \cong BJ, \quad AF \cong BF \\
& \quad \Downarrow \\
& \quad GF \cong JF \\
& \quad \Downarrow \\
& \quad OG \cong OJ, \quad OF \cong OF, \quad GF \cong JF \\
& \quad \Downarrow \\
& \quad \angle GOF \cong \angle JOF \\
& \quad \Downarrow \\
& \quad G - O - J \\
& \quad \Downarrow \\
& \quad \text{RightAngle}(G, O, F).
\end{aligned}$$

The last implication is exactly Euclid's definition of a right angle as two adjacent equal angles formed by a straight line.

## 1.9 The arbitrary transversal: the real formal difficulty

The public conclusion quantifies over *every* line of  $\pi$  through  $O$ . A conventional drawing suggests that such a transversal meets the two connector segments  $AD$  and  $BC$ , but that is false in general. This was the key positional issue in the formal proof.

The correct argument uses two possible connector families:

$$AD/BC \quad \text{or} \quad BD/AC.$$

For a transversal  $q \neq m, n$ , apply Pasch to the triangle  $ABD$ . The line  $q$  enters through  $AB$  at  $O$ , hence it exits either through  $AD$  or through  $BD$ .

This is isolated as

```
hilbert_XI4_transversal_pasch_first_exit
```

Next,

```
hilbert_XI4_opposite_point_on_transversal
```

constructs the point  $J$  opposite  $G$  on  $q$  with

$$G - O - J, \quad OG \cong OJ.$$

Two neutral landing lemmas then prove:

```
hilbert_XI4_first_exit_opposite_lands_on_BC
hilbert_XI4_second_exit_opposite_lands_on_AC
```

so that the opposite point lands on the correct paired connector. The combined package is

```
hilbert_XI4_arbitrary_transversal_connector_package
```

and returns

$$\begin{aligned} G, J \in q, \quad G - O - J, \quad OG \cong OJ, \\ \text{and either} \\ A - G - D, \quad B - J - C, \\ \text{or} \\ B - G - D, \quad A - J - C. \end{aligned}$$

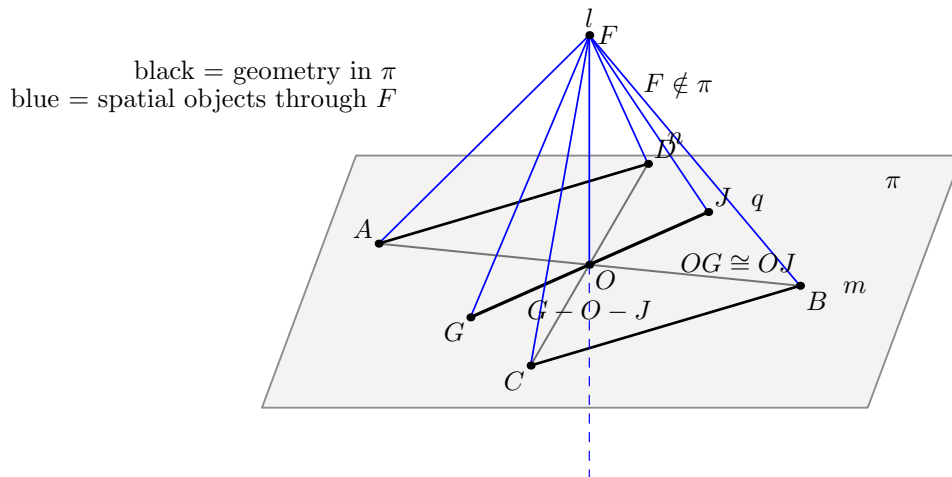


Figure 1.3: Arbitrary-transversal stage. For  $q \subset \pi$  through  $O$ , Pasch selects one of the two connector families. The opposite point  $J$  is constructed with  $G - O - J$  and  $OG \cong OJ$ . The spatial SSS/SAS chain then proves that  $\angle GOF$  is right.

This entire stage uses incidence, order, Pasch, and congruence. It does not use the Euclidean parallel axiom.

## 1.10 Right angle on every transversal

With the connector package available, both branches reduce to the same spatial SSS/SAS scheme. The theorem

```
hilbert_XI4_arbitrary_transversal_right_angle
```

states, in substance:

$$\begin{aligned} q \subset \pi, \quad O \in q, \quad q \neq m, n, \\ F \notin \pi, \quad AF \cong BF, \quad CF \cong DF \end{aligned}$$

imply the existence of  $G \in q$ ,  $G \neq O$ , such that

$$\text{RightAngle}(G, O, F).$$

The two exceptional cases  $q = m$  and  $q = n$  are discharged directly from the hypotheses of XI.4.

## 1.11 Why the point $F$ is outside the plane

The core proof chooses an arbitrary nonvertex point  $F$  on  $l$ . To use spatial noncollinearity cleanly, it proves

$$F \notin \pi.$$

This is the theorem

```
hilbert_XI4_perpendicular_point_off_plane
```

The proof is neutral and conceptually important.

Assume  $F \in \pi$ . Then the line  $l$ , determined by  $O$  and  $F$ , is itself a line of  $\pi$ . Transfer the two ambient perpendicularities to **PlaneGeo** **pi**. Choose nonvertex points  $A \in m$  and  $C \in n$ . Then both

$$\angle FOA \quad \text{and} \quad \angle FOC$$

are right.

The helper

```
hilbert_XI4_two_right_angles_same_first_arm_collinear
```

uses Hilbert III.4 uniqueness to prove that  $A, O, C$  are collinear. Line uniqueness then forces  $m = n$ , contradicting the hypothesis.

No parallel postulate is used.

## 1.12 Closing the universal line–plane statement

The core theorem

```
hilbert_XI4_line_perpendicular_plane_core
```

does the following:

1. choose  $F \neq O$  on  $l$ ;
2. construct the balanced cross  $A, B, C, D$  in  $\pi$ ;
3. prove  $F \notin \pi$ ;
4. prove  $AF \cong BF$  and  $CF \cong DF$ ;
5. let  $q$  be an arbitrary line of  $\pi$  through  $O$ ;
6. if  $q = m$  or  $q = n$ , use the hypotheses;
7. otherwise invoke the arbitrary-transversal theorem and obtain a point  $G \in q$  such that  $\angle GOF$  is right;
8. construct the witness required by `HilbertLinesPerpendicularAt Geo 1 q 0`.

There is one subtle final representation step. The right angle obtained from the transversal theorem is oriented as

$$\angle GOF.$$

The line-line perpendicularity witness is more convenient with the order

$$\angle FOG.$$

The proof does not use an ambient planar congruence instance to swap the arms. Since  $l$  and  $q$  intersect at  $O$ , they determine an explicit plane  $\rho$ . Inside `PlaneGeo rho` the right angle is swapped using the ordinary planar congruence API, and the result is transported back to the ambient geometry.

This preserves the 3D architecture: all planar congruence reasoning occurs inside an explicit plane slice.

### 1.13 Complete formal DAG

The proof architecture can be summarized as follows.

```

two distinct plane lines m,n through O
+ l perpendicular to m and n at O
-> balanced cross A-O-B, C-O-D
-> AD = BC and angle OAD = angle OBC
-> choose F != O on l
-> F notin pi
-> AF = BF, CF = DF
-> arbitrary q in pi through O
-> if q=m or q=n: hypothesis
-> otherwise Pasch in triangle ABD
-> exit through AD or BD
-> opposite J on q, G-O-J, OG = OJ
-> landing on BC or AC
-> ASA: AG = BJ (or BG = AJ)
-> spatial SSS/SAS: GF = JF
-> spatial SSS: angle GOF = angle JOF
-> G-O-J -> RightAngle GOF
-> l perpendicular q at O
-> forall q in pi through O, l perpendicular q
-> HilbertLinePerpendicularPlaneAt Geo l pi O.

```

## 1.14 Classical dependencies versus Lean dependencies

| Role                            | Euclid                            | Lean reconstruction                                                                  |
|---------------------------------|-----------------------------------|--------------------------------------------------------------------------------------|
| Four equal radial segments      | cut off $AE, EB, CE, ED$<br>equal | opposite-point construction on each of the two lines inside <code>PlaneGeo pi</code> |
| Vertical angles                 | I.15                              | developed vertical-angle and angle-congruence infrastructure                         |
| First triangle congruence       | I.4                               | balanced-cross SAS helper                                                            |
| Transversal equalities          | I.26                              | Pasch, transversal ASA, and the two landing branches                                 |
| Distances from $F$              | I.4 on right triangles            | spatial perpendicular-bisector/equidistance helper                                   |
| Cross-plane triangle comparison | I.8 and I.4                       | general spatial SSS/SAS API through explicit carrier planes                          |
| Arbitrary line in the plane     | “drawn at random”                 | universal $q$ ; cases $q = m$ , $q = n$ , or the Pasch branch                        |
| Line perpendicular to plane     | XI.Def.3                          | the universal line-plane perpendicularity predicate                                  |

The Lean DAG is longer because it must prove all incidence, order, noncollinearity, carrier-plane, and orientation facts that the classical diagram leaves implicit. The geometric argument itself remains Euclidean XI.4.

## 1.15 Axiomatic status

The public theorem assumes the spatial Hilbert hierarchy

```

HilbertIncidence Geo
HilbertPlaneIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo

```

```
HilbertSpaceOrder
HilbertSpaceCongruence
```

No `HilbertEuclideanPlane` instance occurs in the theorem signature. In particular, the proof of XI.4 does not use Hilbert Group IV, the Euclidean parallel axiom.

No continuity or Archimedean axiom is introduced by XI.4.

The final Lean audit is:

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[propept,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The project deliberately retains the inherited `bookZero_nullSegment1` dependency at this stage. XI.4 introduces no new proposition-specific axiom.

## 1.16 Formalization notes

### 1.16.1 The proof is synthetic

There are no coordinates, vectors, scalar products, normal vectors, trigonometric functions, angle measures, or real-valued lengths.

All claims are expressed through:

- incidence of points, lines, and planes;
- strict betweenness;
- same-ray and side-of-line relations;
- segment and angle congruence;
- Hilbert right angles;
- Pasch;
- planar and spatial SAS/SSS/ASA arguments.

### 1.16.2 The point $F$ is not a normal-vector surrogate

The point  $F$  is merely a nonvertex point chosen on the candidate line  $l$ . It is used to form ordinary synthetic triangles such as  $AOF$ ,  $BOF$ ,  $GOF$ , and  $JOF$ . The proof never assigns coordinates to  $F$  and never interprets  $OF$  as a vector.



### 1.16.3 Why PlaneGeo is essential

Some arguments are intrinsically planar: Pasch, same-side uniqueness of an angle copy, same-ray normalization, and ordinary Hilbert angle construction. Other arguments compare triangles in different planes. The proof therefore alternates explicitly between:

$$\text{ambient 3D geometry} \longleftrightarrow \text{PlaneGeo}(\pi) \text{ or } \text{PlaneGeo}(\rho).$$

This is not implementation noise. It records exactly where Euclid is using plane geometry inside a solid-geometric proof.

### 1.16.4 Hidden positional data

The formal proof makes explicit all of the following facts:

- the two given plane lines are genuinely distinct;
- perpendicularity witnesses are nondegenerate;
- the four balanced points lie on the intended carriers;
- $A - O - B$  and  $C - O - D$  are strict orders;
- the auxiliary point  $F$  is outside  $\pi$ ;
- a general transversal need not meet the fixed pair  $AD, BC$ ;
- Pasch selects one of two connector families;
- the opposite point on the transversal lies on the paired connector;
- every spatial triangle used by SSS or SAS is noncollinear;
- the final arm swap is performed in the explicit plane through  $l$  and the arbitrary transversal.

These facts are precisely the information supplied visually, but not logically, by a conventional diagram.

### 1.16.5 Why the two connector families matter

A tempting but incorrect formalization is to assume that every line through  $O$  meets both segments  $AD$  and  $BC$ . The diagram can make this appear plausible, but it is not true.

The final proof instead analyzes the first exit of the transversal from triangle  $ABD$ . This produces exactly the dichotomy required by the geometry:

$$(AD, BC) \text{ or } (BD, AC).$$

This is one of the most significant proof-design corrections in the XI.4 development.

## 1.17 Reusable infrastructure produced by XI.4

The proposition required one general correction to the 3D interface: line-line perpendicularity now carries an explicit noncollinearity witness, so that distinctness of the carriers is a theorem rather than an external hypothesis.

The general 3D interface also exposes the bridges

```
HilbertLinesPerpendicularAt
hilbert_linesPerpendicularAt_ne
planeGeo_linesPerpendicularAt_iff_ambient
HilbertLinePerpendicularPlaneAt
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
HilbertLinePerpendicularPlaneAt.incidence
```

The remaining declarations beginning with `hilbert_XI4_` are kept in `Proposition11_4.lean` because they package the concrete Euclidean XI.4 configuration.

## 1.18 Dependency summary

Hilbert spatial incidence + order + congruence  
 $\Downarrow$   
 PlaneGeo slices and ambient/plane bridges  
 $\Downarrow$   
 nondegenerate line-line perpendicularity  
 $\Downarrow$   
 balanced cross  $A - O - B, C - O - D$   
 $\Downarrow$   
 $AD \cong BC, \angle OAD \cong \angle OBC$   
 $\Downarrow$   
 $F \in l, F \neq O, F \notin \pi$   
 $\Downarrow$   
 $AF \cong BF, CF \cong DF$   
 $\Downarrow$   
 Pasch for arbitrary  $q \subset \pi$  through  $O$   
 $\Downarrow$   
 two connector branches + opposite point  $J$   
 $\Downarrow$   
 $GF \cong JF, \angle GOF \cong \angle JOF$   
 $\Downarrow$   
 $\text{RightAngle}(G, O, F)$   
 $\Downarrow$   
 $l \perp q$  for every  $q \subset \pi$  through  $O$   
 $\Downarrow$   
`euclid_proposition_11_4.`

## 1.19 Status

|                                     |                                                 |
|-------------------------------------|-------------------------------------------------|
| Production theorem                  | <code>Geometry.euclid_proposition_11_4</code>   |
| Production file                     | <code>CGJteamLab/Proposition11_4.lean</code>    |
| General 3D interface                | <code>CGJteamLab/Hilbert3DInterface.lean</code> |
| Coordinates/vectors                 | no                                              |
| Euclidean parallel axiom (Group IV) | no                                              |
| Continuity                          | no new use                                      |
| New XI.4-specific axiom             | no                                              |
| Full project build                  | clean                                           |
| Axiom audit                         | completed                                       |

## 1.20 Conclusion

Euclid XI.4 is the first major test of the project’s spatial Hilbert architecture. The proposition begins with only two line-line perpendicularities at a common point and must end with a universal statement about every line of a plane through that point.

The formal proof shows that this transition can be reconstructed synthetically. The essential tools are not coordinates or vector orthogonality, but the same ingredients visible in Euclid’s proof: equal-segment constructions, vertical angles, triangle congruence, an arbitrary transversal, and the definition of a perpendicular to a plane.

Lean exposes two pieces of structure that the classical diagram conceals. First, the arbitrary transversal requires a genuine Pasch argument with two connector families. Second, the repeated uses of Book I congruence in solid geometry require an explicit mechanism for comparing triangles carried by different planes.

Once these points are made explicit, the final statement is exactly Euclid XI.Def.3: the candidate line is perpendicular to every line of the given plane through the common point. In that sense the proof is not a modern replacement of XI.4 but a detailed synthetic reconstruction of its logical content.

## Chapter 2

# Proposition XI.5

### 1. What Proposition XI.5 says

Proposition XI.5 is a statement about the geometry of lines through one point in three-dimensional space.

Let four lines

$$l, \quad m, \quad n, \quad p$$

pass through the same point  $O$ . Assume that  $l$  is perpendicular to each of the other three lines at  $O$ :

$$l \perp m, \quad l \perp n, \quad l \perp p.$$

Assume moreover that  $m \neq n$ .

Then the conclusion is:

The three lines  $m, n, p$  lie in one plane.

The hypothesis  $m \neq n$  is important in the formal statement: two distinct intersecting lines determine a unique plane. Thus  $m$  and  $n$  can be used to construct the plane in which the proposition claims that  $p$  must also lie.

### 2. Geometric picture

The easiest way to visualize the proposition is to think of  $l$  as a vertical line through  $O$ . Every line through  $O$  perpendicular to  $l$  should then lie in one “horizontal” plane through  $O$ .

The theorem says that this intuition is already forced by synthetic geometry. No coordinates, vectors, scalar products, or equations of planes are needed.

The formal proof therefore has two tasks:

1. construct the candidate plane using two of the perpendicular lines;
2. prove that every third line perpendicular to  $l$  at  $O$  must lie in that same plane.

### 3. The key role of Proposition XI.4

Proposition XI.4, already formalized, says:

If a line is perpendicular at a point to two distinct lines of a plane through that point, then it is perpendicular to the plane.

This is exactly the tool needed for XI.5.

Because  $m \neq n$  and both pass through  $O$ , they determine a plane; call it  $\pi$ . Since

$$l \perp m \quad \text{and} \quad l \perp n,$$

Proposition XI.4 gives

$$l \perp \pi \quad \text{at } O.$$

This already tells us something much stronger than the original two line–line perpendicularities:  $l$  is now perpendicular to *every* line of  $\pi$  through  $O$ .

### 4. Why the third perpendicular must lie in the same plane

We now have to prove that  $p \subset \pi$ .

Suppose, for contradiction, that

$$p \not\subset \pi.$$

Since  $l$  and  $p$  are distinct intersecting lines, they determine another plane; call it  $\sigma$ :

$$l \subset \sigma, \quad p \subset \sigma.$$

The planes  $\pi$  and  $\sigma$  are distinct. Indeed, if they were equal, then  $p \subset \sigma$  would imply  $p \subset \pi$ , contrary to our assumption.

Both planes contain  $O$ , so their intersection is a line through  $O$ . Denote this line by  $q$ :

$$q = \pi \cap \sigma.$$

Thus

$$q \subset \pi, \quad q \subset \sigma, \quad O \in q.$$

But  $l \perp \pi$ . Therefore every line of  $\pi$  through  $O$  is perpendicular to  $l$ , and in particular

$$l \perp q.$$

We already had

$$l \perp p.$$

Hence, inside the plane  $\sigma$ , the two lines  $p$  and  $q$  both pass through  $O$  and are both perpendicular to  $l$ .

## 5. The decisive second use of XI.4

At this point one might try to invoke a separate theorem saying that through a point of a plane there is only one perpendicular to a given line. That would work, but it is not necessary.

Assume instead that

$$p \neq q.$$

Both  $p$  and  $q$  lie in  $\sigma$ , and

$$l \perp p, \quad l \perp q.$$

Therefore Proposition XI.4 can be applied a second time, now in the plane  $\sigma$ . It gives

$$l \perp \sigma.$$

But by construction

$$l \subset \sigma.$$

This is impossible. A line perpendicular to a plane cannot itself lie in that plane: otherwise the definition of perpendicularity to a plane would require the line to be perpendicular to itself.

Consequently

$$p = q.$$

Since  $q \subset \pi$ , we obtain

$$p \subset \pi,$$

contradicting the assumption  $p \not\subset \pi$ .

Therefore  $m, n, p$  are coplanar.

## 6. Proof skeleton

The whole argument can be summarized as follows:

$$\begin{aligned} m, n \text{ distinct through } O &\implies \text{a plane } \pi \text{ containing } m, n, \\ l \perp m, l \perp n &\xRightarrow{\text{XI.4}} l \perp \pi, \\ p \not\subset \pi &\implies \text{a plane } \sigma \text{ containing } l, p, \\ \pi \cap \sigma = q &\implies q \subset \pi, q \subset \sigma, \\ l \perp \pi &\implies l \perp q, \\ p \neq q &\xRightarrow{\text{XI.4}} l \perp \sigma, \\ l \subset \sigma &\implies \perp. \end{aligned}$$

Hence  $p = q \subset \pi$ .

## 7. Why this proof is useful for the formal reconstruction

The proof used in the Lean development is deliberately not a literal transcription of Euclid's historical proof.

The guiding principle of the reconstruction is different: once a synthetic theorem has already been proved in the current interface, it may be reused directly if this shortens the later argument without introducing stronger axioms.

Here XI.4 is already available. Applying it twice gives a particularly compact proof of XI.5.

The argument remains entirely synthetic. In particular, it uses no:

- coordinates;
- vectors;
- scalar products;
- trigonometric formulas;
- metric equations of planes.

Only incidence, planes, line intersection, and perpendicularity are used.

## 8. Lean statement

The production theorem is:

```
theorem euclid_proposition_11_5
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [_HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (l m n p : Geo.Line)
  (O : Geo.Point)
  (hmn : Ne m n)
  (hPerpM :
    HilbertLinesPerpendicularAt Geo l m O)
  (hPerpN :
    HilbertLinesPerpendicularAt Geo l n O)
  (hPerpP :
    HilbertLinesPerpendicularAt Geo l p O) :
  exists pi : S.Plane,
    HilbertLineInPlane Geo m pi /\
    HilbertLineInPlane Geo n pi /\
    HilbertLineInPlane Geo p pi
```

The formal conclusion does not merely introduce an abstract “coplanar” predicate. It gives an explicit witness plane  $\pi$  containing all three lines.

## 9. Formal proof architecture

The main library components used by the proof are the following.

### 9.1. Plane through two intersecting lines

The theorem

`hilbert_plane_through_two_intersecting_lines`

constructs  $\pi$  from  $m, n, O$ , and later constructs  $\sigma$  from  $l, p, O$ .

### 9.2. Proposition XI.4

The production theorem

`euclid_proposition_11_4`

is used twice:

1. from  $l \perp m$  and  $l \perp n$ , to obtain  $l \perp \pi$ ;
2. under the temporary assumption  $p \neq q$ , from  $l \perp p$  and  $l \perp q$ , to obtain  $l \perp \sigma$ .

### 9.3. Intersection line of two planes

The theorem

`hilbert_plane_intersection_line`

constructs the line  $q = \pi \cap \sigma$  through  $O$ .

### 9.4. Unpacking line–plane perpendicularity

From  $l \perp \pi$ , the theorem

`HilbertLinePerpendicularPlaneAt.perpendicular_to_line`

gives  $l \perp q$ , because  $q \subset \pi$  and  $O \in q$ .

### 9.5. A line perpendicular to a plane cannot lie in it

A small helper theorem was introduced:

`hilbert_linePerpendicularPlaneAt_not_in_plane`



Its proof is immediate from the definition. If  $l \subset \pi$  and  $l \perp \pi$ , then the line–plane perpendicularity predicate applied to the line  $l$  itself yields

$$l \perp l,$$

contradicting the nondegeneracy of line–line perpendicularity.

## 10. Dependency structure

The conceptual dependency graph is very small:

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}}.$$

More explicitly,

$$\begin{array}{c} \text{3D incidence} \\ \text{plane through intersecting lines} \\ \text{intersection line of two planes} \\ \text{XI.4} \end{array} \implies \text{XI.5}.$$

No new spatial axiom was introduced for XI.5.

## 11. Axiom audit

The Lean audit command

```
#print axioms Geometry.euclid_proposition_11_5
```

returns:

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

This is the same axiom profile as Proposition XI.4.

In particular, XI.5 introduces no additional primitive axiom of three-dimensional geometry.

The local Book Zero axiom

```
Geometry.bookZero_nullSegment1
```

is inherited through the existing planar geometric layer and is not a new assumption specific to XI.5.

## 12. Why XI.5 matters for XI.6

Proposition XI.6 states that two lines perpendicular to the same plane are parallel.

Suppose

$$l \perp \pi \text{ at } B, \quad m \perp \pi \text{ at } D, \quad B \neq D.$$

The line  $BD$  lies in  $\pi$ . Hence

$$l \perp BD, \quad m \perp BD.$$

The difficulty is not the planar fact that two perpendiculars to the same line are parallel. The real three-dimensional difficulty is first to show that  $l$  and  $m$  lie in a common plane.

XI.5 supplies precisely the missing coplanarity mechanism. Thus its formalization is not merely an historical detour: it provides the structural bridge needed for XI.6.

## 13. Status

The production file is:

`CGJteamLab/Proposition11_5.lean`

The audit file is:

`CGJteamLab/Proposition11_5_audit.lean`

The theorem compiles, and the axiom audit is:

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

Thus Proposition XI.5 is formally closed and ready to serve as a dependency of Proposition XI.6.

## Appendix A

# Euclid in Heath's Edition: Source Audit

### A.1 Scope

This appendix records the Book XI source audit against Euclid's *Elements* in Heath's edition at the current checkpoint, Proposition XI.4. The audit records the classical statement, construction, and proposition dependencies. Formal Lean-specific obligations are documented in the proposition chapter rather than attributed to the classical source.

### A.2 Proposition XI.4 in Heath's edition

The proposition states that if a straight line is set up at right angles to each of two straight lines at their point of intersection, then it is also at right angles to the plane through those two lines.

The classical construction chooses equal segments on the two given lines through the common point and then introduces an arbitrary straight line of the plane through that point. The proof reduces the required line-plane perpendicularity to a right-angle statement for this arbitrary line.

The proposition dependencies used in the argument are drawn from Book I triangle congruence and angle theory. In the notation of the formal reconstruction, the proof pattern is summarized

by

$$\begin{array}{c}
A - O - B, \quad C - O - D, \quad OA \cong OB, \quad OC \cong OD \\
\Downarrow \\
\text{first congruent-triangle comparison} \\
\Downarrow \\
\text{connector equalities} \\
\Downarrow \\
G - O - J, \quad OG \cong OJ \\
\Downarrow \\
FG \cong FJ \\
\Downarrow \\
\angle GOF \cong \angle JOF \\
\Downarrow \\
\text{RightAngle}(G, O, F) \\
\Downarrow \\
l \perp \pi.
\end{array}$$

### A.3 Comparison with the Lean reconstruction

The Lean theorem preserves this synthetic proof pattern. The additional formal obligations are:

- explicit construction of the balanced cross;
- explicit treatment of both possible connector families for an arbitrary transversal;
- Pasch-based proofs of the required intersections and orders;
- proof that the chosen nonvertex point  $F$  on the candidate perpendicular lies outside the plane;
- transport between ambient spatial congruence and induced plane geometry.

These are formal expansions of incidence, order, and congruence data that are implicit in the classical diagram.

### A.4 Checkpoint XI.4

At this checkpoint no coordinate, vector, scalar-product, trigonometric, or numerical-angle reformulation is used. The public Lean theorem has the same geometric conclusion as Euclid XI.4: perpendicularity to two distinct intersecting lines of a plane implies perpendicularity to every line of that plane through the common point.

## Appendix B

# Hilbert: Spatial Axiomatic Architecture

### B.1 Scope

This appendix records the Hilbert-side axiomatic audit for Euclid Book XI at Proposition XI.4. The relevant question is which classes of incidence, order, and congruence are used by the formal theorem and whether the Euclidean parallel axiom or a continuity principle enters.

### B.2 Spatial layers in the formal development

The project separates the ambient spatial structure into

```
HilbertSpacePrimitive  
HilbertSpaceIncidence  
HilbertSpaceOrder  
HilbertSpaceCongruence
```

and obtains ordinary planar Hilbert geometry on each fixed plane through

```
PlanePoint  
PlaneLine  
PlaneGeo
```

together with explicit bridges for incidence, betweenness, congruence, same rays, angles, and right angles.

No global

```
HilbertCongruence Geo  
HilbertOrder Geo
```

instances are installed on the ambient three-space. Planar order and planar angle arguments are carried out in a specified `PlaneGeo`; spatial congruence transports data between the required

plane slices.

### B.3 Perpendicularity predicates

For line–line perpendicularity the general interface is

```
HilbertLinesPerpendicularAt
```

and includes an explicit noncollinearity witness for the two arms of the right angle. Hence line distinctness is derived by

```
hilbert_linesPerpendicularAt_ne
```

rather than supplied separately.

Line–plane perpendicularity is represented by

```
HilbertLinePerpendicularPlaneAt
```

which requires the line to be perpendicular, at the common point, to every line of the plane through that point.

### B.4 Axiomatic status of XI.4

The public theorem assumes the spatial incidence/order/congruence hierarchy but no

```
HilbertEuclideanPlane Geo
```

instance. Thus the Euclidean parallel axiom, Hilbert Group IV, is not a hypothesis of XI.4 in the current formal hierarchy.

The axiom audit reported by Lean is

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[proptext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The inherited `bookZero_nullSegment1` dependency is retained at the current checkpoint. No proposition-specific geometric axiom is introduced by XI.4.

**B.5 Checkpoint XI.4**

The formal proof closes in the spatial incidence/order/congruence layer. No Group IV assumption and no new continuity assumption occur in the public theorem.

## Appendix C

# Borsuk–Szmielw: Axiomatic Layers and Spatial Geometry

### C.1 Scope

This appendix records the Book XI audit against Karol Borsuk and Wanda Szmielw, *Foundations of Geometry*, at Proposition XI.4. Their primitive language includes points, lines, planes, betweenness, and equidistance. Their axioms are organized into incidence, order, congruence, continuity, and the Euclidean axiom.

### C.2 Congruence results relevant to XI.4

The Book I audit already identifies the triangle-congruence results used in the XI.4 argument within the pre-Euclidean congruence layer. In particular:

- the SSS theorem corresponding to Euclid I.8 occurs as Borsuk–Szmielw Theorem 21;
- right-angle and perpendicular constructions belong to their synthetic congruence development and are not formulated by numerical angle measure.

The formal XI.4 proof likewise treats SAS/SSS and right-angle transport as congruence infrastructure rather than as consequences of the Euclidean parallel axiom.

### C.3 Spatial geometry

Borsuk–Szmielw develop three-dimensional geometry with points, lines, planes, perpendicular lines, perpendicular projection, and later an absolute coordinate system in space. In that spatial development a line perpendicular to a plane is part of the geometric language before coordinates are introduced as a later representation.

At the current checkpoint no theorem number has been identified in the audited source whose statement is exactly Euclid XI.4. Accordingly the Lean theorem is not attributed here to a specific Borsuk–Szmielw theorem.



## C.4 Axiomatic comparison

For XI.4 the comparison is:

$$\begin{array}{c}
 \text{incidence and order in planes} \\
 + \\
 \text{synthetic segment and angle congruence} \\
 + \\
 \text{spatial incidence/congruence transport} \\
 \Downarrow \\
 \text{line-plane perpendicularity.}
 \end{array}$$

The formal proof does not use the Euclidean parallel axiom. This agrees with the placement of the triangle-congruence ingredients below the Euclidean layer in the Borsuk–Szmielw organization.

## C.5 Checkpoint XI.4

The current audit records no exact Borsuk–Szmielw counterpart for XI.4 itself. Exact theorem-number correspondence should be added only if it is located in the source.

# Bibliography

- [1] Euclid, *The Thirteen Books of Euclid's Elements*, edited and translated by Sir Thomas L. Heath, 3 vols., Dover Publications, New York, 1956. (Reprint of the second edition, Cambridge University Press, 1926.)
- [2] David Hilbert, *Grundlagen der Geometrie*, B. G. Teubner, Leipzig, 1899.
- [3] Karol Borsuk and Wanda Szmielew, *Foundations of Geometry: Euclidean and Bolyai-Lobachevskian Geometry, Projective Geometry*, North-Holland Publishing Company, Amsterdam, 1960.